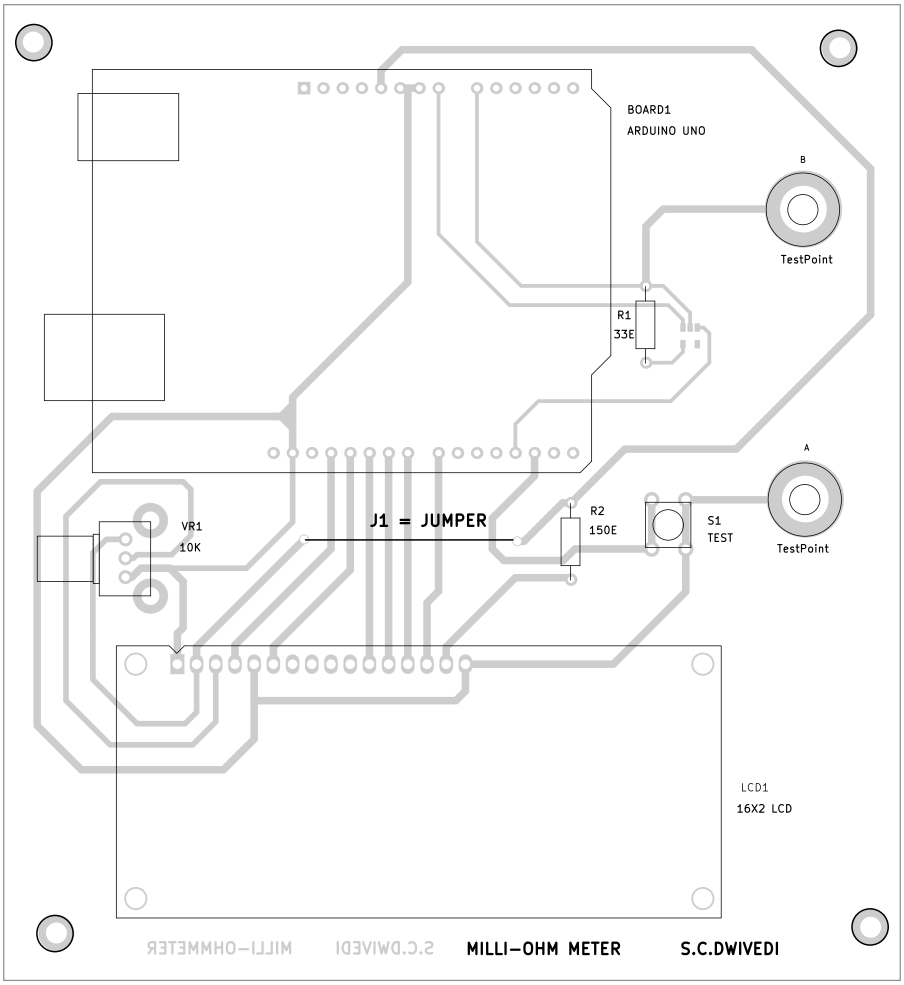

Fig. 4: Component layout of the PCB

Arduino IDE is used to compile and upload the program.

Construction and testing

An actual-size, single-side PCB for the milli-ohm meter is shown in Fig. 3 and its component layout in Fig. 4. Before using the circuit, do not forget to upload the source code mR_Meter.ino by burning it into the microprocessor in Arduino Uno board.

After assembling the circuit on PCB, enclose it in a suitable box. The test switch S1 should be fixed at a convenient position in the cabinet so that it can be used easily.

The Arduino Uno can be powered by a 9V battery or a 9V, 500mA adaptor. Use a suitable connector for test points A and B for connecting the test resistance to be measured.

EFY notes. 1. This circuit measures low resitances of 0.1-ohm to 1-ohm only. There may be an error/deviation in the measurement due to parasitic elements in the PCB.

2. Any other suitable low drop-out regulator (LDO) can be used in place of MIC5219 for IC1.

3. A 1.5V Zener diode can be connected across the analogue pin and ground to bypass the current source if no resistance is connected when the LDO is enabled by the Arduino. EFY

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